

Force and Motion

A Science A-Z Physical Series

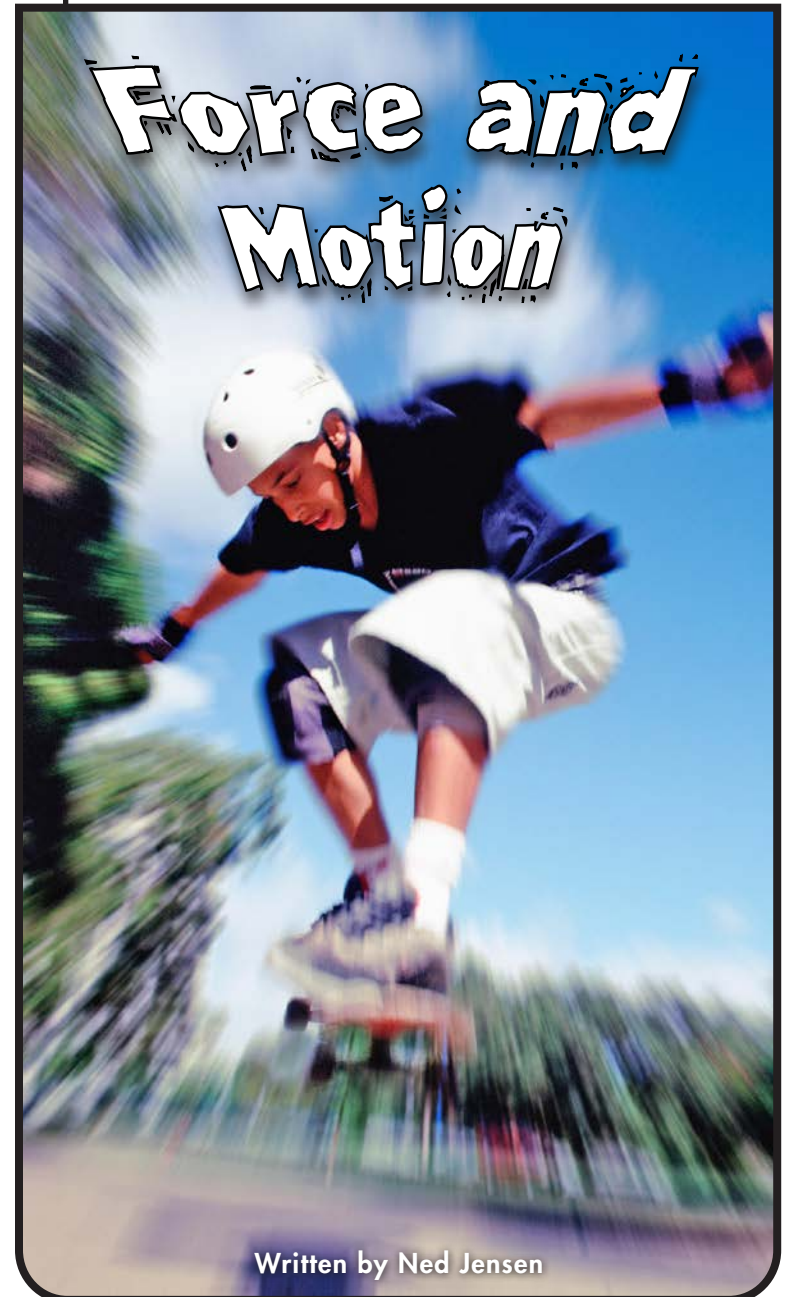
Word Count: 2,299



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Written by Ned Jensen

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KEY ELEMENTS USED IN THIS BOOK

The Big Idea: Force and motion are fundamental to all matter in the universe. A force is anything that can push or pull on an object. Forces influence objects that are at rest or that are already in motion. Isaac Newton's three laws of motion involve inertia, mass, velocity, and momentum. Key forces include gravity, friction, and magnetism. A force is required to do work, and generating a force requires energy. Energy can be stored as potential energy, or it can have kinetic energy—the energy of motion. Energy can also be converted and exchanged through energy transfer. Objects move in predictable ways. By learning about force and motion, we come to understand how using forces can produce motions that allow us to be safe and to enjoy ourselves.

Key words: attract, chemical energy, direction, distance, electricity, electromagnetism, energy, energy transfer, engine, force, friction, generator, gravity, heat energy, inertia, kinetic energy, law, light energy, lines of force, magnetic field, magnetism, mass, momentum, motion, potential energy, reaction, repel, rest, sound energy, speed, velocity, weight, work

Key comprehension skills: Cause and Effect

Other suitable comprehension skills: Classify information; compare and contrast; elements of a genre; identify facts; interpret charts, graphs, and diagrams; main idea and details

Key reading strategy: Visualize

Other suitable reading strategies: Ask and answer questions; connect to prior knowledge; retell; summarize

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Introduction

Each word above describes how objects can move. Some things move in a straight line, such as a ball rolled to you along the ground. When you kick the ball into the air, it flies in a curved path and then bounces on the ground until it stops. A merry-go-round revolves in circles, while lightning strikes in a jagged path.

Motion is everywhere, and it's vital to life. Think of how your heart's movement keeps blood flowing throughout your body! In this book, you will learn what makes things move, stop, change **direction**, and change **speed**.

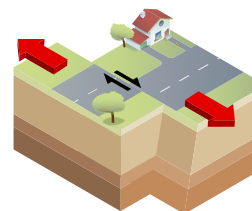
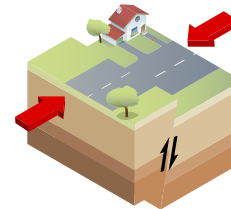
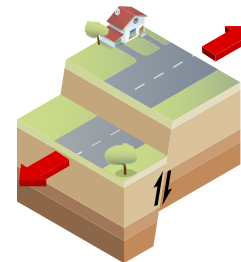
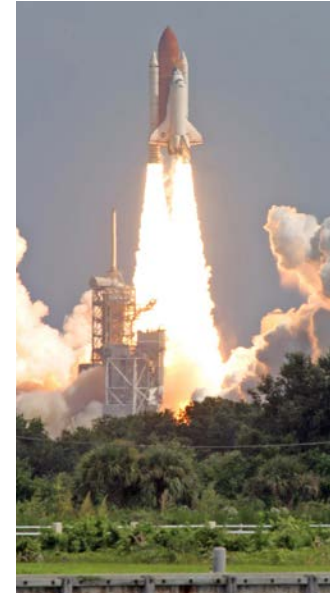


Motion Needs a Force

In order to make something move, a **force** is required. Lifting, pushing, and pulling are all types of forces, and those forces come from countless sources. Sometimes a person can provide the force. If you lift a stack of books, push a broken-down car, or pull on a rope in a tug-of-war game, you are the source of a force.



Machines can also produce a force to move objects. The power of a huge engine can produce enough force to propel a massive rocket into space. A car's smaller engine makes its wheels spin, moving the vehicle down a road. Motors in cake mixers, fans, and blenders use electrical force to make things move.

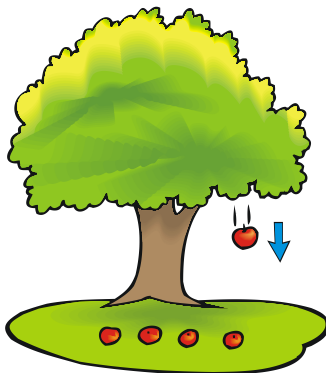


Nature can act as a force, too. For example, breezes make leaves dance in the treetops. Ocean waves make boats bob up and down. Forces deep underground can cause earthquakes that make the ground tremble and destroy buildings, bridges, and highways.

Earth's crust is made up of moving plates. These plates can shift in various ways. The force of this shifting causes earthquakes.

The Laws of Motion

Much of what we know about motion comes from experiments and observations made by scientists hundreds of years ago. The Englishman Isaac Newton is especially famous in this regard. You may have heard of him as the man who was sitting under a tree when an apple fell and hit him on the head. The story goes on to tell how this event led to his theory about **gravity**, the invisible force that causes things to fall toward the ground.



Actually, Isaac wasn't hit on the head by a falling apple—that's a myth. But over time, he observed that objects always fell down toward the ground. These observations, combined with his curiosity, drove him to study gravity.

SIR ISAAC NEWTON (1642–1727)

Isaac Newton was born on Christmas Day in Woolsthorpe, England. He lived on a farm with his grandmother. He had little interest in becoming a farmer and chose instead to study math and science at Trinity College, which is part of the University of Cambridge in England. At age 27, Newton conducted experiments with light and was the first to suggest that light was actually made up of all the colors. He eventually became one of the most famous scientists of all time!



Newton's First Law of Motion

Isaac Newton is best known for writing three laws of motion that explain how all things move. The first part of Newton's first law says that an object at rest will remain at rest unless a force acts upon it. For example, your bike will stay parked where it is unless someone or something moves it.

The second part of the first law states that an object in motion will continue in motion at the same speed and in the same direction unless a force acts upon it. So if your bike is rolling, it will continue moving just as it is until you pedal faster, coast, brake, or turn. If a force, such as a headwind, pushes from the opposite direction of the bike's motion, it will slow the bike or stop it completely. If a force, such as side wind, is applied to one side of the moving bike, the bike will change direction.



Why is it important to wear a seat belt? Think about Newton's first law of motion.

Word Wise

The word *law* often means a rule made by a government that people must obey. In science, a law is a statement about results that always occur under certain circumstances.

Whether an object is at rest or in motion, it has **inertia**—the tendency to keep doing what it is already doing. Unless a force overcomes the inertia of an object at rest, it will not move. Similarly, an object in motion will keep moving unless a force stops it. So inertia is the reason an object at rest stays at rest and an object in motion stays in motion—until a force acts on it. Newton’s first law of motion is sometimes called the law of inertia.



An object has inertia when it is at rest, and it also has inertia when in motion. In either case, a force is required to overcome inertia and cause motion to start, stop, or change.

Newton’s Second Law of Motion

Why is it easier to move a small rock than a boulder? The boulder has much more **mass**. Scientists define mass as the amount of matter in an object. Newton’s second law of motion explains that an object’s motion depends on its mass and the amount of force being used against that mass.

Since a large boulder has much more mass than a small rock, much more force is needed to overcome the boulder’s inertia. Much less force is required to make small rocks move, stop, or change direction.





This train has a large mass and a high speed, so it has a lot of momentum. It would take a very strong force to slow or stop this train!

While mass is important to motion, speed matters, too. Speed is a measure of how much **distance** is traveled in a particular amount of time. The faster an object is moving, the more force is needed to stop it.

Think about a runaway locomotive speeding down a track. It has both a lot of mass and a lot of speed. A great deal of force will be required to overcome the inertia of the train's motion and make it stop. This law of motion explains why trains cannot stop suddenly.

Scientists measure the amount of inertia as **momentum**—the force that keeps an object moving.

Word Wise

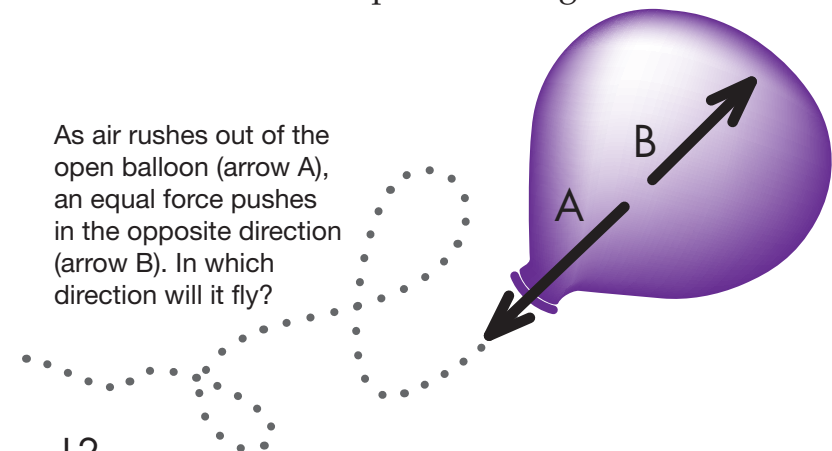
Velocity describes an object's speed in a certain direction. Scientists calculate momentum by using this formula:

$$\text{MASS} \times \text{VELOCITY} = \text{MOMENTUM}$$

Newton's Third Law of Motion

Newton's third law of motion says that for every force or action, there is an equal and opposite reaction. Think of forces in pairs. When you push or pull on an object, the object sends an equal push or pull on you in the opposite direction. So, if you lift a 9 kg (20 lb.) box, the box is pulling downward with an equal force of 9 kg (20 lbs.) in the opposite direction.

This third law of motion also explains why a balloon full of air goes flying when you let it go without tying it. As air rushes out the open end, an opposite force pushes on the other end of the balloon, making it fly. Similarly, a rocket's upward flight is an opposite reaction to the hot exhaust shooting downward from the powerful engines.



As air rushes out of the open balloon (arrow A), an equal force pushes in the opposite direction (arrow B). In which direction will it fly?

Types of Forces

Earlier, you read about lifting, pushing, and pulling, as well as machines and nature all being sources of forces. Read on to learn about several additional important forces.

Gravity

Isaac Newton observed that all objects in the universe attract other objects due to gravity. This law is known as the universal law of gravity. The strength of gravitational pull between two objects depends on their masses and distance. Greater masses have a stronger pull. The longer the distances between them, the weaker the pull.

MASS, DISTANCE, AND GRAVITY

In which situation is the gravitational pull strongest?

In which situation is the gravitational pull weakest?

A  • two small masses • short distance	B  • two small masses • greater distance
C  • two large masses • short distance	D  • two large masses • greater distance

Strongest: C; Weakest: B

Understanding the effects of mass and distance on gravity will help you understand why objects fall to Earth rather than toward each other. Imagine you are standing next to a large building. Both you and the building have mass, so you both have gravitational pull. The building has much more mass, so it pulls you more than you pull it.



However, both you and the building are standing on Earth, which is far more massive than either you or the building. Earth's gravity is so much stronger that it has much more gravitational pull on you and the building than you and the building have on each other. The gravitational pull between you and the building goes unnoticed. This is true of all objects on Earth.

On Earth, an object's mass is measured as **weight**. The greater the mass of an object, the more the force of Earth's gravity pulls on it and, therefore, the more it weighs.



If you were in space, twice as far from the center of Earth as you are now, the gravitational pull would be one-quarter as strong. You would still have the same mass, but you would weigh one-quarter as much as you do on Earth.

Earth seems huge to us, but the Sun has much more mass than Earth or anything else in our solar system. As a result, it has enough gravitational pull to keep all the planets from flying off into space. Instead, they all orbit the Sun.

Remember, both mass and distance are important aspects of gravity. So while the Sun has a greater gravitational pull than Earth, the distance between your gravitational center and Earth's gravitational center is much less than the distance between your gravitational center and the Sun's gravitational center. So you remain pulled toward Earth. That's a good thing! If Earth were too close to the Sun, you would be pulled toward it.

Friction

Friction is an invisible force that slows down moving things. Friction changes motion energy to heat energy. As an object gains heat energy due to friction, it has less motion energy, so it slows down.

The most familiar kind of friction is *sliding friction*. When you rub your hands together firmly, they create sliding friction, which produces heat energy. That's why your skin warms up.

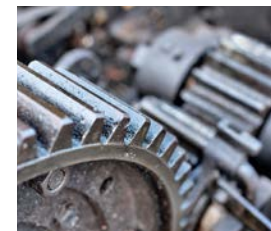
A rolling bowling ball experiences another kind of friction—*rolling friction*. Rolling friction will slow down a moving object, but not as quickly as sliding



friction. If you shove a heavy box across the floor, it will soon slow down due to sliding friction. But if you place rollers under the box, it will move farther when shoved since rolling reduces friction.



Rough surfaces exert more friction against each other than smooth surfaces do. Putting a lubricant such as oil or grease between two surfaces will reduce friction.



Another kind of friction, *fluid friction*, occurs when an object moves in contact with a fluid, such as water or air. It is easier to move your finger through water than honey because honey creates more fluid friction than water. Engineers design airplanes to reduce the drag caused by fluid friction with the air. Fluid friction can help objects move more easily, such as a puck floating on air in an air hockey game. However, objects experiencing fluid friction will slow down and eventually come to a stop.



Think About It

Imagine that you wanted to race your friends down a snowy hill. How could thinking about friction help you win the race?

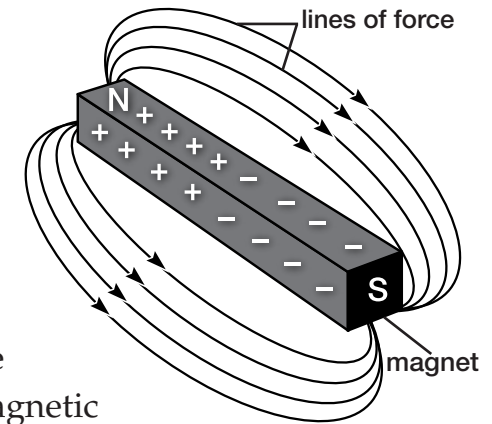


Magnetism

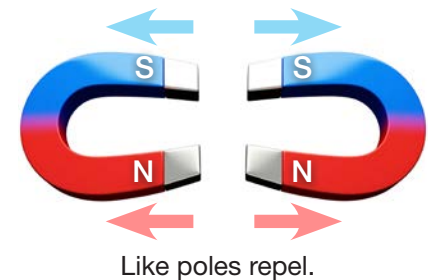
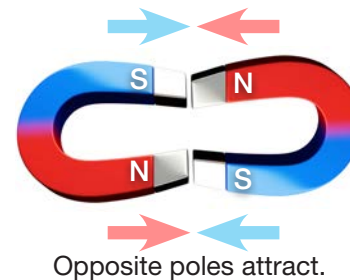
Magnetism is a force that can push or pull objects made of certain metals, such as iron. Magnets also push and pull each other.

Magnetism can act on objects without even touching them. How? A magnet is surrounded by invisible *lines of force* that enter and leave at opposite ends, or *poles*, of the magnet. When a magnetic object enters the magnetic field, it is attracted to the magnet.

Magnetic Field Around a Magnet



Each magnet has a north and south pole. A magnet's force is strongest near its poles. Two opposite poles *attract* each other, but two of the same poles *repel*, or push each other away.

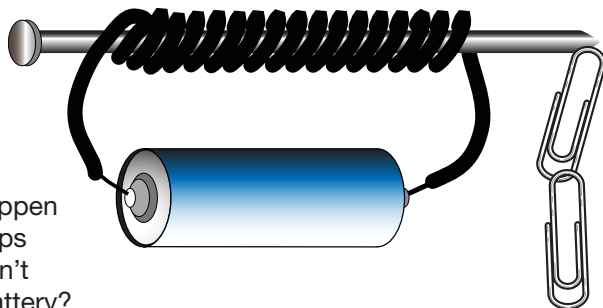




Generators like these use magnets to make electricity flow through wires.

Magnets can make electricity move through a wire. Large machines called *generators* use magnets to produce electricity in power plants.

On the other hand, electricity can turn some metals into magnets. Electricity and magnetism are actually both part of a single force called **electromagnetism**.



What would happen to the paper clips if the wire weren't touching the battery?

Force, Motion, and Work

In science, you do **work** when you move something. If you push against a wall all day long and it doesn't move, you have applied a force, but you have not done work.

The amount of work you do depends on two things: the distance the object moves and the amount of force it takes to move it. The farther you move an object, the more work you do. And the more force you apply to a moving object, the more work you have done. Scientists measure work by multiplying the distance an object moves by the force used to move it:

$$\text{FORCE} \times \text{DISTANCE} = \text{WORK}$$

4 blocks, no baby	most work 4 blocks, with baby
least work 2 blocks, no baby	2 blocks, with baby

It takes less work to push an empty stroller than a stroller with a baby in it because an empty stroller has less mass. Also, pushing the stroller four city blocks requires twice as much work as pushing it only two blocks.

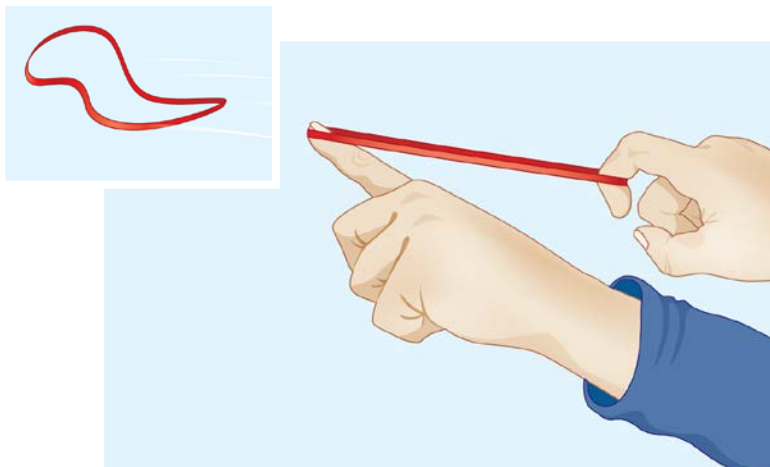
Energy

If you've ever pushed a stroller or lifted a heavy box, you know that it takes energy to do work—sometimes a lot of energy. In fact, energy is defined as the ability to do work. The more energy you have, the more work you can do.



Potential and Kinetic Energy

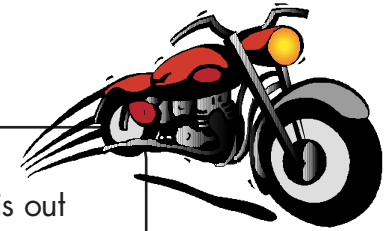
Energy comes in two basic types: potential and kinetic. Take a rubber band, for example. When you pull it back, it has **potential energy**; the energy is stored and ready to use. The farther back you pull the rubber band, the more potential energy it has. When you let it go, you convert that potential energy into **kinetic energy**, and the rubber band flies across the room. Energy in motion is kinetic energy.



Can you guess which type of energy a tank full of gasoline has? Fuels such as gasoline have potential energy; they are energy sources. Burning gas to make a car move converts that potential energy into kinetic energy. A full tank has twice as much potential energy as a tank that is half full.



Why can't a motorcycle that is out of gas be ridden? Think about potential and kinetic energy.



The food you eat also has potential energy. Plants get energy from the Sun, and animals get energy from eating plants and other animals. When you eat and digest food, you store up energy. That energy makes your heart pump and makes your lungs expand and contract. Stored energy also makes your muscles work, allowing you to run, jump, speak, write, and smile.



Energy Transfer

Energy can be changed, or *converted*, from one kind to another. Energy can also move from one object or place to another. Converting and moving energy is called **energy transfer**.

Solar energy can turn into stored energy in plants through photosynthesis. The stored energy in foods can be transferred to your muscles and converted into motion energy in your arms. Then the motion energy in your arms can be transferred to moving drumsticks. The moving sticks can make a drum vibrate, creating sound energy.



Chemical energy in batteries can make a toy robot move. Some of the energy might be converted into sound energy to make sounds. The energy may also be converted into light energy to operate colorful lights. What other examples of energy transfer have you read about?



Conclusion

You've learned that it takes force to make things move or stop and that force can affect the direction and speed of a moving object. You read about Isaac Newton's laws of motion and how the mass of an object is related to the force needed to make it move. Simple pushes and pulls as well as lifting are examples of force. Other types of force include gravity, friction, and magnetism. People can exert a force, as can machines and nature.

You've also learned that work is done when a force moves an object over a distance. It takes energy to do work. This energy is either potential (stored) or kinetic (motion). Energy can be transferred by moving it from one object to another or by changing it from one type of energy to another.

Observe all the motion in your world. Which forces are making things move, and which forces are affecting those moving objects? Soon you'll discover that forces and motion are all around you.



Glossary

direction	the way or course toward which something moves or faces (p. 4)
distance	the amount of space between things (p. 11)
electromagnetism	the combined force of electricity and magnetism (p. 19)
energy transfer	the movement of energy from one object to another or the change of energy from one form to another (p. 23)
force	the strength or energy that moves an object (p. 5)
friction	a force that slows down moving things (p. 16)
gravity	the force that pulls things toward the center of Earth or any other object that has mass (p. 7)
inertia	the tendency of an object to resist change in the direction or speed of its motion (p. 9)
kinetic energy	the energy that a moving body has because of its motion (p. 21)
magnetism	a force that pushes and pulls certain metals (p. 18)

mass	the amount of matter, measured on Earth by its weight (p. 10)
momentum	the strength or force that keeps something moving (p. 11)
motion	the act of going from one place to another; movement (p. 4)
potential energy	the energy a body has because of its position, electrical charge, or structure; stored energy (p. 21)
speed	the rate of movement (p. 4)
weight	how heavy something is, determined by the pull of gravity on the object's mass (p. 14)
work	the act of using force to move something over a certain distance (p. 20)

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